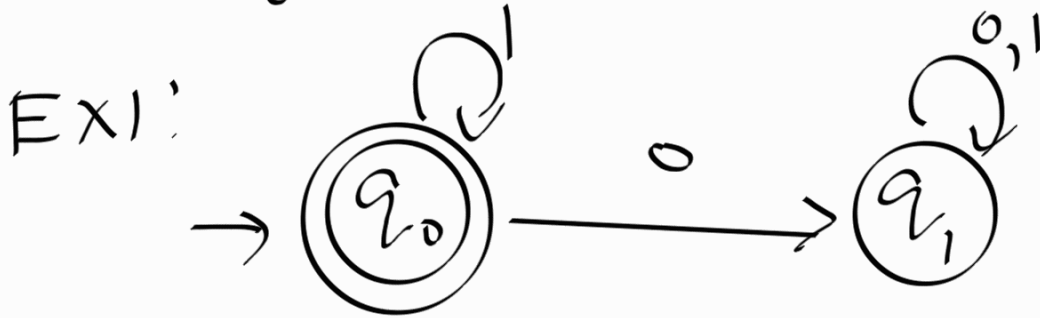


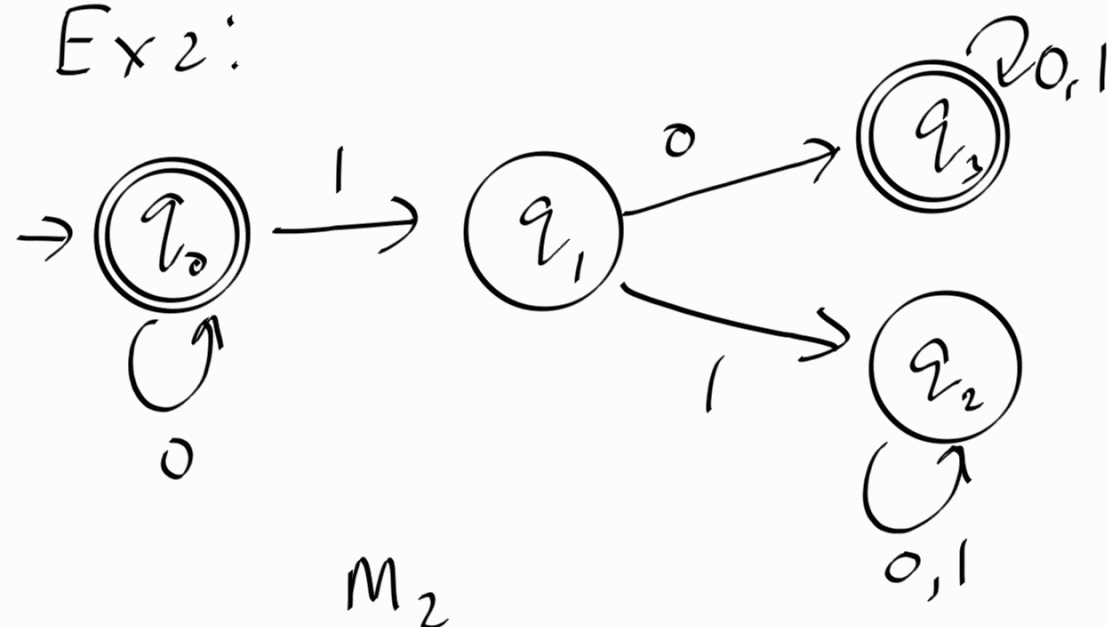
Let us try to identify
languages accepted by the
following machines. $\Sigma = \{0, 1\}$



$$L(M_1) = \{1^n \mid n \geq 0\}$$

$$L(M_1) = 1^*$$

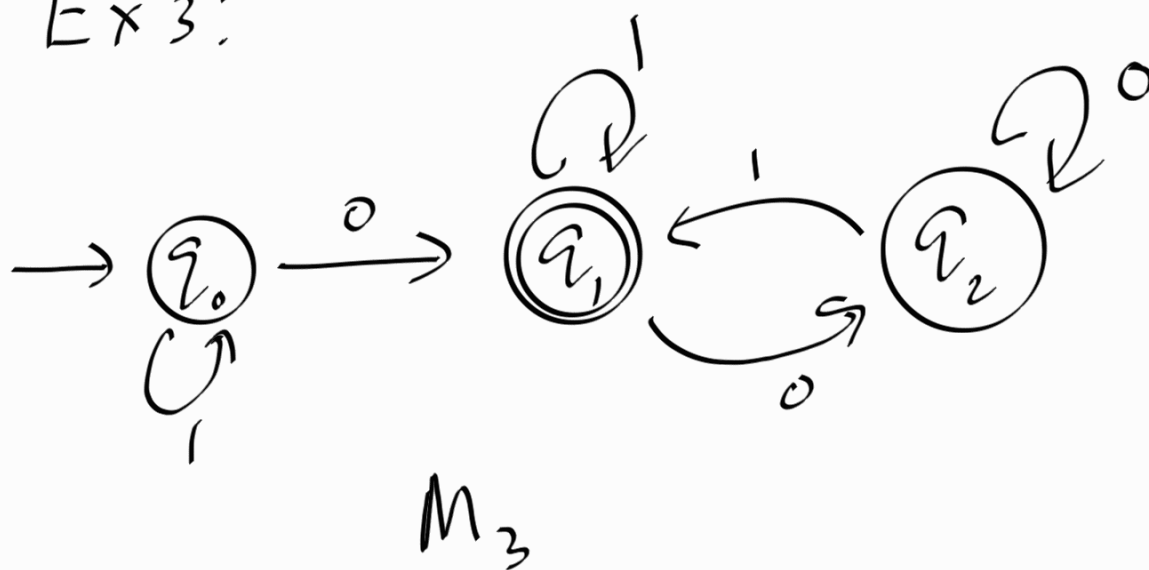
Ex 2:



100

$$L(M_2) = 0^* \cup 0^* 1 0 (0|1)^*$$

Ex 3:



$$L(M_3) = 1^* 0 (1^* | (0 0^* 1)^*)^*$$

Definition 1.16

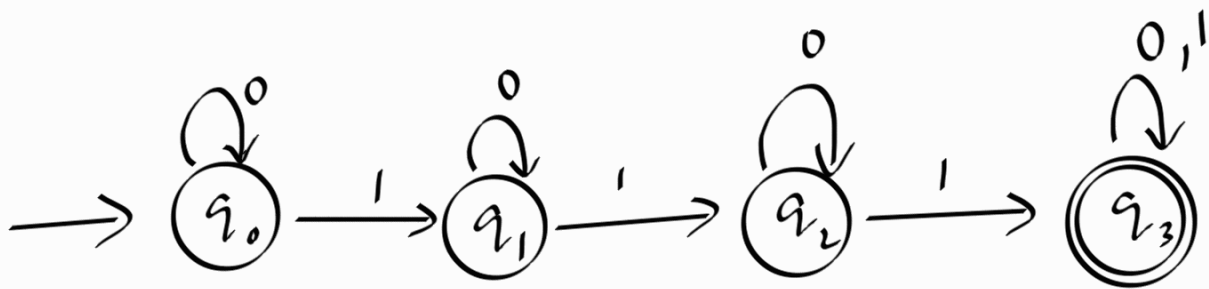
Regular Language:

A language is called regular if some DFA accepts the language.

How to design finite automata?

Ex: Construct a DFA that accepts language
 $A = \{w \mid w \text{ contains at least three 1's}\}$

$$\Sigma = \{0, 1\}$$



Ex 2: $A = \{ w \mid w \text{ starts and ends with the same symbol} \}$

$$\Sigma = \{ a, b \}$$

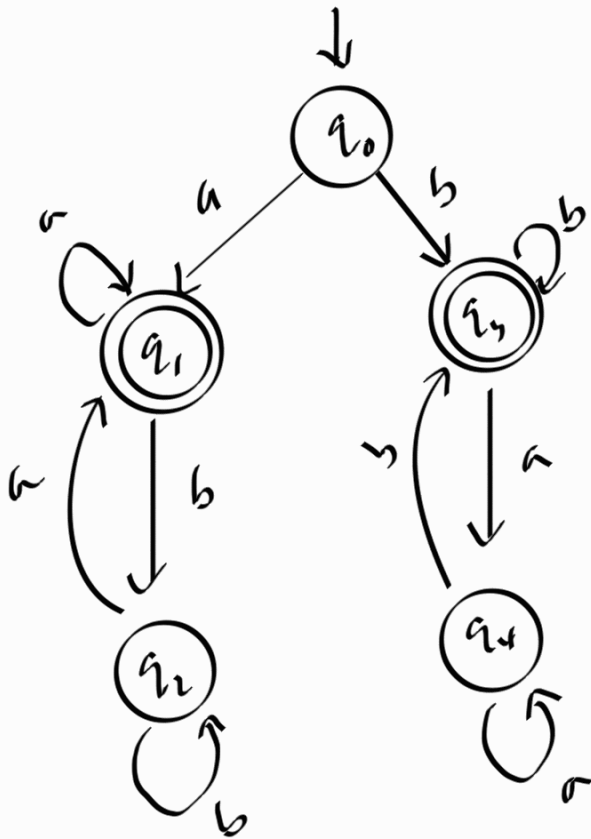
$\varepsilon = ""$ x

✓ a, b ✓

✓ aa, bb ✓

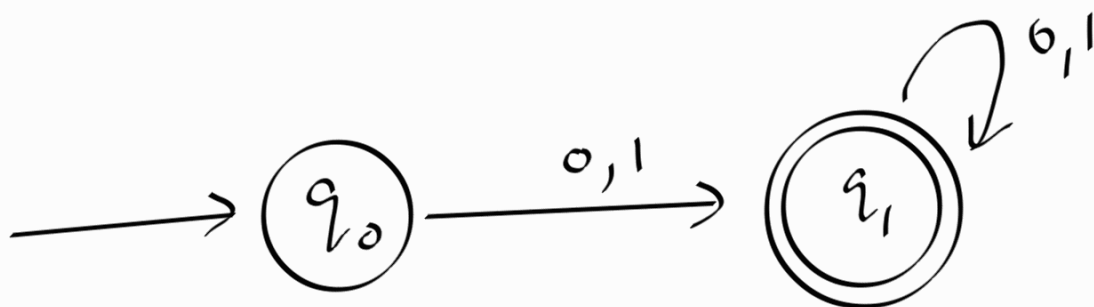
ab x

ba x



Ex 3: DFA that accepts every string except the empty string.

$$\Sigma = \{0, 1\}$$



Regular operations

Let A, B be ^{Regular} languages

1. Union: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

2. Concatenation: $A \circ B = \{xy \mid x \in A \wedge y \in B\}$

3. Star: $A^* = \{x_1 x_2 \dots x_k \mid k \geq 0 \text{ and each } x_i \in A\}$

$$\Sigma = \{a, b, c, \dots, z\}$$

$$A = \{\text{good}, \text{bad}\} \quad B = \{\text{dog}, \text{cat}\}$$

$$A \cup B = \{\text{good}, \text{bad}, \text{dog}, \text{cat}\}$$

$$A \circ B = \{\text{goodcat}, \text{good dog}, \text{bad dog}, \text{badcat}\}$$

$$A^* = \{\epsilon, \text{bad bad}, \text{bad good bad}, \dots\}$$

$$A \circ B \neq B \circ A \quad \text{when } A \neq B$$

Side Note

If $x \in \mathbb{N}$, $y \in \mathbb{N}$, $x + y \in \mathbb{N}$

we say that natural numbers are closed under addition operator

Natural numbers are not closed under division operator

$7 \in \mathbb{N}$, $8 \in \mathbb{N}$, but $8/7 \notin \mathbb{N}$

Theorem 1.25

The class of regular languages are closed under the union operation.

In other words,

If A_1, A_2 are regular languages
so is $A_1 \cup A_2$

Proof $\Rightarrow$

Assume A_1, A_2 are regular and M_1, M_2 recognize A_1 and A_2 respectively.

By definition, let M_1 be a DFA that recognizes A_1 , i.e.,

$$M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$$

Similarly, $M_2 = (Q_2, \Sigma_2, \delta_2, q_2, F_2)$
recognizes A_2 .

We construct $M = (Q, \Sigma, \delta, q_0, F)$

$$1. Q = Q_1 \times Q_2$$

$$2. \Sigma = \Sigma_1 \cup \Sigma_2$$

$$3. \delta : Q \times \Sigma \rightarrow Q$$

$$\delta((r_1, r_2), a) = (\delta_1(r_1, a), \delta_2(r_2, a))$$

$$4. q_0 = (q_1, q_2)$$

$$5. F = \{(r_1, r_2) : r_1 \in F_1 \text{ or } r_2 \in F_2\}$$

